Software Requirements Specification (SRS)

1. Introduction

1.1 Purpose: This document specifies the requirements for a web-based puzzle platform. The system shall allow users to practice puzzle games and compete in multiplayer challenges.

1.2 Scope: The system shall provide single-player puzzles, multiplayer competitions, and performance tracking.

1.3 Definitions: A 'Puzzle' is a logic-based game. A 'Match' is a multiplayer session. A 'Leaderboard' ranks users based on performance.

2. Overall Description

2.1 Product Perspective: The system shall be a web application with a backend API and real-time capabilities.

2.2 User Classes: The system shall support Guest Users, Registered Users, and Admin users.

2.3 Assumptions: Users must have internet access. Real-time features require a stable connection.

3. Functional Requirements

3.1 Authentication: The system shall allow users to register, log in, and log out. The system must securely store user credentials.

3.2 Puzzle System: The system shall allow users to select and solve puzzles. The system must provide immediate feedback on answers.

3.3 Multiplayer Mode: The system shall match users into games. The system must synchronise timers and track puzzle completion in real time.

3.4 Scoring and Leaderboards: The system shall calculate scores based on speed and accuracy. The system must maintain global and weekly leaderboards.

3.5 Timer-Based Challenges: The system shall allow users to select a time duration. The system must end sessions when time expires.

3.6 User Profiles: The system shall store user statistics and match history.

4. Non-Functional Requirements

4.1 Performance: The system shall load pages within 2 seconds. Real-time interactions must have latency below 200ms.

4.2 Scalability: The system shall support increasing numbers of concurrent users.

4.3 Security: The system must implement secure authentication and input validation.

4.4 Usability: The system shall provide an intuitive and mobile-friendly interface.

5. System Architecture

The system shall consist of a frontend (React), backend (Node.js or Django), database (PostgreSQL), and real-time communication via WebSockets.

6. Data Requirements

The system shall store user data, match data, and puzzle data including identifiers, timestamps, and results.

7. Future Enhancements

The system may include ranked matchmaking, AI opponents, and user-generated puzzles.